

KARAN SRIVASTAVA

Department of Mathematics
University of Wisconsin-Madison
480 Lincoln Dr.
Madison, WI, 53706

Phone: (217) 904-9350
Email: ksrivastava090198@gmail.com
Website: karansrivastava.com
LinkedIn: [ksrivastava1](https://www.linkedin.com/in/ksrivastava1)
Github: [ksrivastava1](https://github.com/ksrivastava1)
September 2025

EDUCATION

University of Wisconsin-Madison	<i>Madison, WI</i>
MA, PhD in Mathematics, Minor in Computer Science	<i>2020–present</i>
University of Illinois at Urbana-Champaign	<i>Urbana-Champaign, IL</i>
BS Mathematics	<i>2016–2020</i>
Magna Cum Laude, Highest Distinction in Mathematics	
Independent University of Moscow, Math in Moscow Program	<i>Moscow, Russia</i>
Graduate Courses in Pure Mathematics	<i>Spring 2019</i>

INDUSTRY EXPERIENCE SUMMARY

IBM Research, Research Intern *May 2025 – Aug 2025*

- Genetically trained a large language model (LLM) to evolve heuristics for vehicle routing, achieving 40% faster convergence on TSP instances vs. standard ALNS, with strong generalization to out-of-distribution problems.
- Developed a differential vanishing component analysis framework to discover higher-order dynamical systems from data, outperforming state-of-the-art symbolic regression via numerical linear algebra and optimization.

IBM Research, Exploratory Math Sciences Sr. Research Intern *May 2024 – Aug 2024*

- Developed a geometric and optimization framework for discovering scientific formulae from background theory and data, achieving a 98% speed improvement and 1000x memory reduction over state-of-the-art.
- Created a framework for discovering missing polynomial background axioms using computational algebraic geometry with 95% accuracy.
- Designed and released the first benchmark dataset for symbolic regression models incorporating background theory (polynomials and differential equations).

PUBLICATIONS AND PREPRINTS

K. Srivastava, S. Dash, R. Cory-Wright, B. Trager, L. Horesh. *AI Noether: Bridging the Gap Between Scientific Laws Derived by AI Systems and Canonical Knowledge via Abductive Inference*. arXiv:2509.23004, 2025.

K. Srivastava, J. Lenchner, J. Goncalves, L. Horesh. *SynPAT: A System for Generating Synthetic Physical Theories with Data*. arXiv:2505.00878, 2025.

J. S. Ellenberg, C. S. Fraser-Taliente, T. R. Harvey, K. Srivastava, A. V. Sutherland. *Generative Modeling for Mathematical Discovery*. arXiv:2503.11061, 2025. Submitted to *Advances in Theoretical and Mathematical Physics*.

K. Srivastava, D. Pimentel-Alarcon. *A Perturbation Bound on the Subspace Estimator from Canonical Projections*. IEEE International Symposium on Information Theory (ISIT), 2022.

CONFERENCES

IEEE International Symposium on Information Theory

Presented an accepted publication

Espoo, Finland

June 2022

TALKS

Computational algebraic methods for abductively inferring axioms to explain a phenomenon

Alan Turing Institute

London, England

June 2025

Symbolic Regression and Polynomial Optimization in Scientific DIsccovery

Graduate Applied Math Seminar

Madison, WI

Dec 2024

Automated discovery of Erdos collinear free subsets with RL

Institute for the Foundations of Data Science

Madison, WI

March 2024

Reinforcement Learning for Generating Isosceles Free Subsets of a Lattice.

Graduate Applied Math Seminar

Madison, WI

Jan 2024

Advancing problems in searching subsets of large integer lattices with RL

Specialty Exam Talk

Madison, WI

Jan 2024

Generating large isosceles-free subsets of finite integer lattices with RL

Institute for the Foundations of Data Science

Madison, WI

Nov 2023

Using Reinforcement Learning for Generating Useful Combinatorial Data

IBM Research at Almaden

San Jose, CA

July 2023

Subspace estimation with Noise

Wisconsin Institute for Discovery, Lightning Talks

Madison, WI

July 2022

A Perturbation Bound on the Subspace Estimator from Canonical Projections

International Symposium on Information Theory

Espoo, Finland

June 2022

An almost Impossible puzzle and group theory

AMS Student Chapter Seminar

Madison, WI

June 2022

Why people say "I can't do math"

Presented an Ignite Talk at Park City Math Institute's Summer School.

You can watch it [here](#)

Park City, UT

July 2019

AWARDS AND HONORS

IFDS Research Assistantship

\$25,000 Research Grant

Institute for Foundations of Data Science

Aug 2023 – May 2024

Campus-wide Exceptional Service Award

Awarded to 3 of 2300 Teaching Assistants campus-wide

University of Wisconsin

2022–2023

Departmental Exceptional Service Award

Awarded to 2 of 120 Math Department Teaching Assistants

University of Wisconsin

2022–2023

Exceptional Teaching Award

Awarded to TAs demonstrating excellence in teaching

University of Wisconsin

2020–2022

Edmund J. James Scholar
Awarded to top 15% of undergraduates campus-wide

University of Illinois
2017–2020

Dean's List
Awarded every semester

University of Illinois
2016–2020

ORGANIZATIONS AND OUTREACH

Madison Experimental Mathematics Lab
Graduate Coordinator

University of Wisconsin
2022-present

Directed Reading Program
Organizer, Mentor

University of Wisconsin
Fall 2021-2024

Undergraduate Mentor Program
Co-Founder, Organizer, Mentor

University of Wisconsin
2022-2024

Graduate Peer Mentor Program
Mentor and Organizer

University of Wisconsin
Fall 2022, 2023-2024

Madison Math Circle
Organizer

University of Wisconsin
2021-2022

Grad Student Visit Day Panel
Graduate Coordinator

University of Wisconsin
Fall 2021, Fall 2022

TEACHING EXPERIENCE

SL Math

Summer School: Principled Scientific Discovery with Formal Methods

IBM

Lecturer and Co-organizer

June 2025

University of Wisconsin

Math 345: Linear Algebra and Optimization

University of Wisconsin

Co-designed course and teaching Assistant*

Spring 2025

Math 240: Introduction to Discrete Mathematics

University of Wisconsin

Teaching Assistant*

Summer 2022, 2023

Math 222: Calculus and Analytical Geometry II

University of Wisconsin

Head Teaching Assistant*

Summer 2021, Spring 2022

Math 211: Calculus

University of Wisconsin

Head Teaching Assistant*

Fall (2021,22,24,25), Spring (21,22)

Math 221: Calculus and Analytical Geometry I

University of Wisconsin

Teaching Assistant*

Fall 2020

*Awarded superior teaching assistant rating by the Department of Mathematics

OTHER RESEARCH EXPERIENCE

Illinois Gometry Lab

Undergrad Research Program

Advisor: Dominic Culver, Topic: Weierstrass equations of Elliptic Curves

Advisor: Susan Tolman, Topic: Sphere packings on Symplectic Manifolds

University of Illinois

Spring 2020

Fall 2018

SELECTED PROJECTS

Reinforcement Learning for Tic Tac Toe

July 2023 – Aug 2023

Built a reinforcement learning agent for Tic Tac Toe that achieved perfect play in 10,000+ games without heuristics. [Github Link](#).

Causal Inference and Machine Learning Project, University of Wisconsin

Fall 2022

Designed Support Vector Regression models that uncovered causal relationships in synthetic and real-world data, achieving ~90% accuracy. [Write-up Link](#). [Github Link](#).

Copayment Prediction, Erdős Institute Data Science Bootcamp

Nov 2022 – Jan 2023

Worked in a team of 3 PhDs to train ensemble models on a 13M-row healthcare dataset. Reduced RMSE from ~ \$40 baseline to \$15.30. [Github Link](#).

Cuisine Classification, Erdős Institute Data Science Bootcamp

Summer 2022

Used dimensionality reduction and classification models (Random Forest, Linear SVC, etc.) on the “What’s Cooking?” dataset from Kaggle, achieving ~80% accuracy. [Project Link](#).